

CAMBRIDGE INTERNATIONAL EXAMINATIONS

Cambridge International Advanced Subsidiary and Advanced Level

MARK SCHEME for the May/June 2015 series

9702 PHYSICS

9702/42

Paper 4 (A2 Structured Questions), maximum raw mark 100

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42

- 1 (a) (i) 1. $F = Gm_1m_2/x^2$
 $= (6.67 \times 10^{-11} \times 2.50 \times 5.98 \times 10^{24}) / (6.37 \times 10^6)^2$
 $= 24.6 \text{ N}$ (accept 2 s.f. or more) M1
A1 [2]
2. $F = mx\omega^2$ or $F = mv^2/x$ and $v = \omega x$ (accept x or r for distance)
 $= 2.50 \times 6.37 \times 10^6 \times (2\pi/24 \times 3600)^2$
 $= 0.0842 \text{ N}$ (accept 2 s.f. or more) C1
A1 [2]
- (ii) reading $= 24.575 - 0.0842$ B1
 $= 24.5 \text{ N}$ (accept only 3 s.f.) A1 [2]
- (b) gravitational force provides the centripetal force M1
gravitational force is 'equal' to the centripetal force
(accept $Gm_1m_2/x^2 = mx\omega^2$ or $F_c = F_g$) M1
'weight'/sensation of weight/contact force/reaction force is difference between F_g and F_c which is zero A1 [3]
- 2 (a) mean speed $= 1.44 \times 10^3 \text{ m s}^{-1}$ A1 [1]
- (b) evidence of summing of individual squared speeds C1
mean square speed $= 2.09 \times 10^6 \text{ m}^2 \text{ s}^{-2}$ A1 [2]
- (c) root-mean-square speed $= 1.45 \times 10^3 \text{ m s}^{-1}$ A1 [1]
(allow ECF from (b) but only if arithmetic error)
- 3 (a) (numerically equal to) quantity of heat/(thermal) energy to change state/phase of unit mass M1
at constant temperature A1 [2]
(allow 1/2 for definition restricted to fusion or vaporisation)
- (b) (i) constant gradient/straight line (allow linear/constant slope) B1 [1]
- (ii) $Pt = mL$ or power $= \text{gradient} \times L$ C1
use of gradient of graph M1
(or two points separated by at least 3.5 minutes)
 $110 \times 60 = L \times (372 - 325) \times 10^{-3} / 7.0$
 $L = 9.80 \times 10^5 \text{ J kg}^{-1}$ (accept 2 s.f.) (allow 9.8 to 9.9 rounded to 2 s.f.) A1 [3]
- (iii) some energy/heat is lost to the surroundings or vapour condenses on sides M1
so value is an overestimate A1 [2]
- 4 (a) displacement (directly) proportional to acceleration/force M1
either displacement and acceleration in opposite directions
or acceleration (always) towards a (fixed) point A1 [2]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42

- (b) (i) $\frac{1}{3}\pi$ rad or 1.05 rad (*allow 60° if unit clear*) A1 [1]
- (ii) $a_0 = -\omega^2 x_0$
 $= (-) (2\pi/1.2)^2 \times 0.030$
 $= (-) 0.82 \text{ m s}^{-2}$
(special case: using oscillator P gives $x_0 = 1.7 \text{ cm}$ and $a_0 = 0.47 \text{ m s}^{-1}$ for 1/2) C1
A1 [2]
- (iii) max. energy $\propto x_0^2$
ratio $= 3.0^2/1.7^2$
 $= 3.1$ (*at least 2 s.f.*)
(if has inverse ratio but has stated max. energy $\propto x_0^2$ then allow 1/2) C1
A1 [2]
- (c) graph: straight line through (0,0) with negative gradient
correct end-points (−3.0, +0.82) and (+3.0, −0.82) M1
A1 [2]
- 5 (a) work done bringing/moving per unit positive charge
from infinity (to the point) M1
A1 [2]
- (b) (i) slope/gradient (of the line/graph/tangent)
*(allow dV/dx , but **not** $\Delta V/\Delta x$ or V/x)*
(allow potential gradient)
(negative sign not required) B1 [1]
- (ii) maximum at surface of sphere A or at $x = 0$ (cm)
zero at $x = 6$ (cm)
then increases but in opposite direction
(any mention of attraction max. 2/3) B1
B1
B1 [3]
- (c) (i) M shown between $x = 5.5 \text{ cm}$ and $x = 6.5 \text{ cm}$ B1 [1]
- (ii) 1. $\Delta V = (570 - 230) = 340 \text{ V}$ (*allow 330 V to 340 V*) A1 [1]
2. $q(\Delta)V = \frac{1}{2}mv^2$ **or** change/loss in PE = change/gain in KE **or** $\Delta E_K = \Delta E_P$ B1
- $4.8 \times 10^7 \times 340 = \frac{1}{2}mv^2$ C1
- $v^2 = 3.26 \times 10^{10}$
- $v = 1.8 \times 10^5 \text{ m s}^{-1}$ (*not 1 s.f.*) A1 [3]
- 6 (a) packet/quantum/discrete amount of energy
of electromagnetic energy/radiation/waves M1
A1 [2]
- (b) (i) arrow below axis and pointing to right B1 [1]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42

- (ii) 1. $E = hc/\lambda$
 $= (6.63 \times 10^{-34} \times 3.0 \times 10^8)/(6.80 \times 10^{-12})$
 $= 2.93 \times 10^{-14} \text{ J (accept 2 s.f.)}$ C1
A1 [2]
2. energy of electron $= (3.06 - 2.93) \times 10^{-14}$
 $= 1.3 \times 10^{-15} \text{ J}$ C1
speed $= \sqrt{2E/m}$ C1
 $= 5.4 \times 10^7 \text{ ms}^{-1}$ A1 [3]
- (c) momentum is a vector quantity B1
either must consider momentum in two directions B1
or direction changes so cannot just consider magnitude B1 [2]
- 7 (a) moving magnet gives rise to/causes/induces e.m.f./current in solenoid/coil B1
(induced current) creates field/flux in solenoid that opposes (motion of) magnet B1
work is done/energy is needed to move magnet (into solenoid) B1
(induced) current gives heating effect (in resistor) which comes from the work done B1 [4]
- (b) current in primary coil give rise to (magnetic) flux/field B1
(magnetic) flux/field (in core) is in phase with current (in primary coil) B1
(magnetic) flux threads/links/cuts secondary coil inducing e.m.f. in secondary coil B1
(there **must** be a mention of secondary coil)
e.m.f. induced proportional to rate of change/cutting of flux/field so not in phase B1 [4]
- 8 (a) (i) energy $= 5.75 \times 1.6 \times 10^{-13}$
 $= 9.2 \times 10^{-13} \text{ J}$ A1 [1]
- (ii) number $= 1900/(9.2 \times 10^{-13} \times 0.24)$ C1
 $= 8.6 \times 10^{15} \text{ s}^{-1}$ A1 [2]
- (b) (i) decay constant $= 0.693/(2.8 \times 365 \times 24 \times 3600)$ C1
 $= 7.85 \times 10^{-9} \text{ s}^{-1}$ (allow 7.8 or 7.9 to 2 s.f.) A1 [2]
- (ii) $A = \lambda N$
 $8.6 \times 10^{15} = 7.85 \times 10^{-9} \times N$ C1
 $N = 1.096 \times 10^{24}$ C1
mass $= (1.096 \times 10^{24} \times 236)/(6.02 \times 10^{23})$ M1
 $= 430 \text{ g}$ A1 [4]
- (c) $0.84 = 1.9 \exp(-7.85 \times 10^{-9} t)$ C1
 $t = 1.04 \times 10^8 \text{ s}$
 $= 3.3 \text{ years}$ A1 [2]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42

Section B

- 9 (a) $V_B = 1000 \text{ mV}$ C1
when strained, $V_A = 2000 \times 121.5 / (121.5 + 120.0)$
 $= 1006.2 \text{ mV}$ M1
change = 6.2 mV (*allow 6 mV*) A1 [3]
- (b) (i) 1. resistor between V_{IN} and V^- and V^+ connected to earth B1
resistor between V^- and V_{OUT} B1 [2]
2. P/+ sign shown on earth side of voltmeter B1 [1]
- (ii) ratio of $R_F / R_{IN} = 40$ M1
 R_{IN} between 100Ω and $10 \text{ k}\Omega$ A1 [2]
(*any values must link to the correct resistors on the diagram*)
- 10 (a) product of density (of medium) and speed (of ultrasound) M1
in the medium A1 [2]
- (b) (i) $7.0 \times 10^6 = 1.7 \times 10^3 \times \text{speed}$ C1
speed = $4.12 \times 10^3 \text{ ms}^{-1}$
wavelength = $(4.12 \times 10^3) / (9.0 \times 10^5) \text{ m}$ C1
 $= 4.6 \text{ mm}$ (*2 s.f. minimum*) A1 [3]
- (ii) for air/tissue boundary, $I_R / I \approx 1$ M1
for air/tissue boundary, (almost) complete reflection/no transmission A1
for gel/tissue boundary, $I_R / I = 0.1^2 / 3.1^2$
 $= 1.04 \times 10^{-3}$ (*accept 1 s.f.*) M1
gel enables (almost) complete transmission (into the tissue) A1 [4]
- 11 (a) (i) metal (*allow specific example of a metal*) B1 [1]
- (ii) e.g. provides 'return' for the signal
shields inner core from interference/reduces cross-talk/reduces noise
increased security
(*any two sensible suggestions, 1 each*) B2 [2]
- (b) (i) (gradual) loss of power/intensity/amplitude B1 [1]
- (ii) dB is a log scale B1
either large (range of) numbers are easier to handle (on a log scale)
or compounding attenuations/amplifications is easier B1 [2]
- (c) attenuation = $190 \times 11 \times 10^{-3} = 2.09 \text{ dB}$ C1
 $-2.09 = 10 \lg(P_{OUT} / P_{IN})$ C1
ratio = 0.62 A1 [3]

Page 6	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2015	9702	42

- 12 handset transmits (identification) signal to number of base stations B1
base stations transfers (signal) to cellular exchange B1
(idea of stations needed at least once in first two marking points)
- computer at cellular exchange selects base station with strongest signal B1
computer at cellular exchange selects a carrier frequency for mobile phone B1 [4]
(idea of computer needed at least once in these two marking points)